

R E V I E W

Post-insertion outcomes of ultrasound-guided versus landmark peripheral intravenous catheters: A systematic review and meta-analysis

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ABSTRACT

Background: Ultrasound-guided (USG) peripheral intravenous catheter (PIVC) insertion improves cannulation success, particularly in patients with difficult intravenous access. Whether USG also improves post-insertion outcomes such as catheter failure, dwell time, and complications remains uncertain.

Methods: We conducted a systematic review and meta-analysis of randomized controlled trials and comparative cohort studies comparing USG with landmark-guided PIVC insertion and reporting at least one post-insertion outcome. PubMed, Embase, Cochrane CENTRAL, and CINAHL were searched from January 2000 to March 2026. Risk of bias was assessed using RoB 2 and the Newcastle-Ottawa Scale. Random-effects meta-analysis used the DerSimonian-Laird estimator as the pre-specified primary approach; REML with Hartung-Knapp-Sidik-Jonkman adjustment was performed as a post-hoc sensitivity analysis because few studies contributed to each pooled outcome.

Results: Fifteen studies (6 randomized trials, 9 cohort studies; 78,591 participants) were included in the qualitative synthesis. Catheter failure did not differ significantly between USG and landmark groups ($k = 4$; RR 1.23, 95% CI 1.00–1.51; $p = 0.056$), although the estimate was sensitive to model choice and to exclusion of the largest contributing study. Dwell time and extravasation were not pooled because of substantial heterogeneity ($I^2 = 91.9\%$ and 95.7% , respectively). Infiltration did not differ significantly ($k = 3$; RR 1.30, 95% CI 0.82–2.05). Several studies



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reporting favorable USG results were substantially confounded by systematic differences in catheter length or material between groups.

Conclusions: Current evidence does not demonstrate a consistent post-insertion advantage of USG over landmark PIVC placement. Apparent benefits reported in some studies may be influenced by catheter-specification differences rather than by ultrasound guidance itself. Because the evidence base is small and heterogeneous, clinically meaningful benefit or harm cannot be excluded. Adequately powered trials that control for catheter length, gauge, and material are needed to determine whether USG independently influences post-insertion catheter performance. These findings are constrained by a small, heterogeneous evidence base, catheter-specification confounding in several positive studies, and low-to-very-low GRADE certainty.

Key words: peripheral intravenous catheter, ultrasound-guided, landmark, difficult intravenous access (DIVA), catheter failure, dwell time, extravasation, systematic review, meta-analysis

Introduction

Peripheral intravenous catheter (PIVC) insertion is the most commonly performed invasive procedure in hospitals worldwide, with an estimated one billion catheters placed annually across the globe (1). Despite its ubiquity, PIVC insertion remains a significant source of patient discomfort, procedural delay, and healthcare resource consumption, particularly among patients with difficult intravenous access (DIVA). Over the past two decades, ultrasound-guided (USG) PIVC insertion has emerged as a widely endorsed technique to improve cannulation success, especially when traditional landmark and palpation methods have failed (2, 3).

The evidence supporting USG PIVC insertion has focused predominantly on insertion success metrics, including first-attempt success rates, overall success rates, number of attempts, and time to successful cannulation. Multiple systematic reviews and meta-analyses have demonstrated that USG improves first-attempt success and reduces the number of skin punctures compared with the landmark technique, findings that have driven the incorporation of USG PIVC into clinical practice guidelines and training curricula (4–7), though findings in general pediatric populations have been less consistent (8). However, successful insertion of a PIVC represents only the beginning of the device's clinical lifecycle. Post-insertion

outcomes, including catheter dwell time, catheter failure, phlebitis, infiltration, extravasation, occlusion, dislodgement, and catheter-related infection, are arguably of equal or greater clinical importance, as they determine the functional utility of the catheter and the need for replacement procedures.

Despite the clinical relevance of post-insertion outcomes, no prior systematic review has specifically synthesized the evidence comparing post-insertion performance of USG versus landmark PIVCs. The assumption that improved insertion success will translate into improved catheter longevity and fewer complications has not been rigorously tested. Theoretical concerns exist that USG catheters, which often access deeper veins with standard-length catheters, may be at increased risk of mechanical complications such as infiltration and dislodgement due to a shorter intravenous catheter segment within the vessel lumen (9, 10).

We therefore conducted a systematic review and meta-analysis to evaluate post-insertion outcomes of USG compared with landmark PIVC insertion. Our primary outcomes were catheter failure and dwell time. Secondary outcomes included infiltration, extravasation, phlebitis, occlusion, dislodgement, and catheter-related infection. Rather than presuming a direction of effect, we evaluated whether post-insertion catheter performance differs between USG and landmark placement.

Methods

Search strategy

This systematic review and meta-analysis was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines and was prospectively registered in PROSPERO (CRD420261354170) (see Supplementary Material S1). A comprehensive literature search was performed across four electronic databases: PubMed (n = 530), Embase (n = 984), Cochrane CENTRAL (n = 68), and CINAHL (n = 50), yielding 1,632 total records. The search spanned from 1 January 2000 to 27 March 2026 and was restricted to English-language publications. Search terms combined controlled vocabulary and free-text keywords related to three conceptual domains: (1) peripheral intravenous catheterization (e.g., “peripheral venous catheter,” “intravenous cannulation,” “PIVC”); (2) ultrasound guidance (e.g., “ultrasonography,” “ultrasound-guided,” “echography”); and (3) post-insertion outcomes (e.g., “catheter failure,” “dwell time,” “phlebitis,” “infiltration,” “extravasation,” “complications”). Reference lists of included studies and relevant systematic reviews were hand-searched to identify additional eligible records (see Supplementary Material S2).

Study selection

After removal of 273 duplicates, 1,359 unique records underwent title and abstract screening by two reviewers (C.C.C. and Y.L.W.) independently in Rayyan (Rayyan Systems Inc., Cambridge, MA, USA). Overall percent agreement was 95.8% (Cohen's kappa = 0.330). Both reviewers concurrently excluded 1,287 records; 15 were both-reviewer includes; 57 records were single-reviewer includes (4 PI-only, 53 second-reviewer-only). Single-reviewer-include records underwent pre-specified second-pass title/abstract adjudication by the primary reviewer against the PROSPERO PICO; 53 were judged outside scope at this stage with documented reasons (first-attempt insertion outcome; midline/long-peripheral catheter; single-arm; unpublished protocol), and 4 advanced to full-text for confirmation. In total, 19 records advanced to full-text retrieval.

All 19 records advancing to full-text consideration were retrieved and screened. Full-text eligibility was assessed independently by two reviewers (T.A.L. and Y.L.W.), with the senior author (C.C.C.) adjudicating; after consensus discussion, 15 studies met all inclusion criteria and were included in the final analysis. Four reports were excluded at the full-text stage: (1) Malik et al., single-arm study without a comparator group (11); (2) one Avelar et al. report (Rev Esc Enferm USP, 2013), available only in Portuguese with no reliable translation (12); (3) one oncology-nurses report, unpublished at time of search; (4) Kleidon et al. (BJN, 2023) (13), a trial-protocol-only publication of the same EPIC trial without post-insertion outcome data, excluded as an overlapping report of the single included EPIC primary publication (Kleidon et al., JAMA Pediatrics, 2025) (14). A further overlapping EPIC conference abstract was also identified and excluded earlier at the title/abstract screening stage. Two studies (Desai et al., 2018 (15); Nishizawa et al., 2020 (16)) were not available through initial institutional access but were subsequently retrieved via interlibrary loan and included in the final analysis (see Supplementary Material S3). The English-language report of the Avelar trial (Journal of Infusion Nursing, 2015 (5)) met all eligibility criteria and was included; the 2013 Portuguese article (12) is a companion report of the same trial (ClinicalTrials.gov NCT00930254) rather than a separate study.

Eligibility criteria

Studies were eligible for inclusion if they: (1) directly compared USG with landmark or palpation-guided PIVC insertion; (2) reported at least one post-insertion outcome, including dwell time, catheter failure, phlebitis, infiltration, occlusion, dislodgement, or catheter-related infection; (3) employed a randomized controlled trial (RCT) or comparative cohort design; (4) enrolled human participants; and (5) were published in English between 2000 and 2026.

Data extraction

Data were extracted independently by two reviewers using a standardized form. Extracted variables included: study design, country, setting, sample size by group,

participant age and population characteristics (adult vs. pediatric), DIVA status, catheter specifications (length, gauge, material), follow-up duration, and all reported post-insertion outcomes with their effect estimates and measures of variance. For studies reporting median and interquartile range, these were recorded as presented. Authors were not contacted for unpublished data.

Risk of bias assessment

Risk of bias for RCTs was assessed using the Cochrane Risk of Bias 2 (RoB 2) tool (Figure 1A) across five domains: randomization process, deviations from intended interventions, missing outcome data, measurement of the outcome, and selection of the reported result. Each RCT was classified as low risk, some concerns, or high risk of bias. Observational studies were evaluated using the Newcastle-Ottawa Scale (NOS) (Figure 1B), with scores ranging from 0 to 9 stars across three domains: selection, comparability, and outcome (Figure 1C). The certainty of evidence for each outcome was rated using the GRADE approach (considering risk of bias, inconsistency, indirectness, imprecision, and publication bias), with observational studies starting at low certainty; ratings are summarized in Table 1 (17).

Statistical analysis

Random-effects meta-analyses were conducted using the DerSimonian-Laird (DL) method for outcomes reported by two or more studies with comparable definitions and populations. Risk ratios (RR) with 95% confidence intervals (CI) were calculated for dichotomous outcomes; mean differences (MD) with 95% CI were calculated for continuous outcomes. For studies reporting hazard ratios (HR), these were pooled separately. Statistical heterogeneity was assessed using the I^2 statistic, with values of 25%, 50%, and 75% representing low, moderate, and high heterogeneity, respectively. Outcomes with I^2 exceeding 75% were not pooled and were described narratively.

Prespecified subgroup analyses were conducted by age group (adult vs. pediatric) and study design (RCT vs. cohort). Sensitivity analyses were performed by restricting the analysis to RCTs only and by leave-one-out analysis to assess the influence of individual

studies. Publication bias could not be formally assessed (e.g., by funnel plot or Egger's test) because fewer than 10 studies contributed to any single pooled analysis. Studies identified as having major confounding by systematic differences in catheter type or length between groups were excluded from the primary meta-analysis and reported separately.

All analyses were performed using R (version 4.5.3) with the meta (18) and metafor (19) packages. A two-sided p -value < 0.05 was considered statistically significant.

The DerSimonian-Laird (DL) estimator was the pre-specified primary approach in our PROSPERO protocol, reflecting standard practice in systematic reviews at the time of registration. After the initial analytic run, we recognized that with only 2–4 studies contributing to each outcome, DL standard errors are known to be anti-conservative and the resulting confidence intervals can be narrower than warranted (20). We therefore added a post-hoc sensitivity analysis using the restricted maximum likelihood (REML) estimator and the Hartung-Knapp-Sidik-Jonkman (HKSJ) confidence interval adjustment, which is the currently recommended approach for small- k random-effects meta-analyses. We retain DL as the pre-specified primary estimator (per PROSPERO) but explicitly flag that, where DL and HKSJ estimates straddle the conventional $\alpha = 0.05$ threshold (as occurs for catheter failure: DL, $p = 0.056$ vs HKSJ, $p = 0.047$), interpretation should not hinge on either side of that threshold and the overall pattern of evidence should drive clinical inference rather than a single estimator's significance.

Results

Study characteristics

The search identified 1,632 records across four databases. After removing 273 duplicates, 1,359 unique records were screened by title and abstract; 1,287 were excluded by both reviewers, and 72 met a union-inclusion threshold of which 53 single-second-reviewer includes were excluded at second-pass title/abstract adjudication by the primary reviewer (per pre-specified consensus discussion procedure). The remaining 19

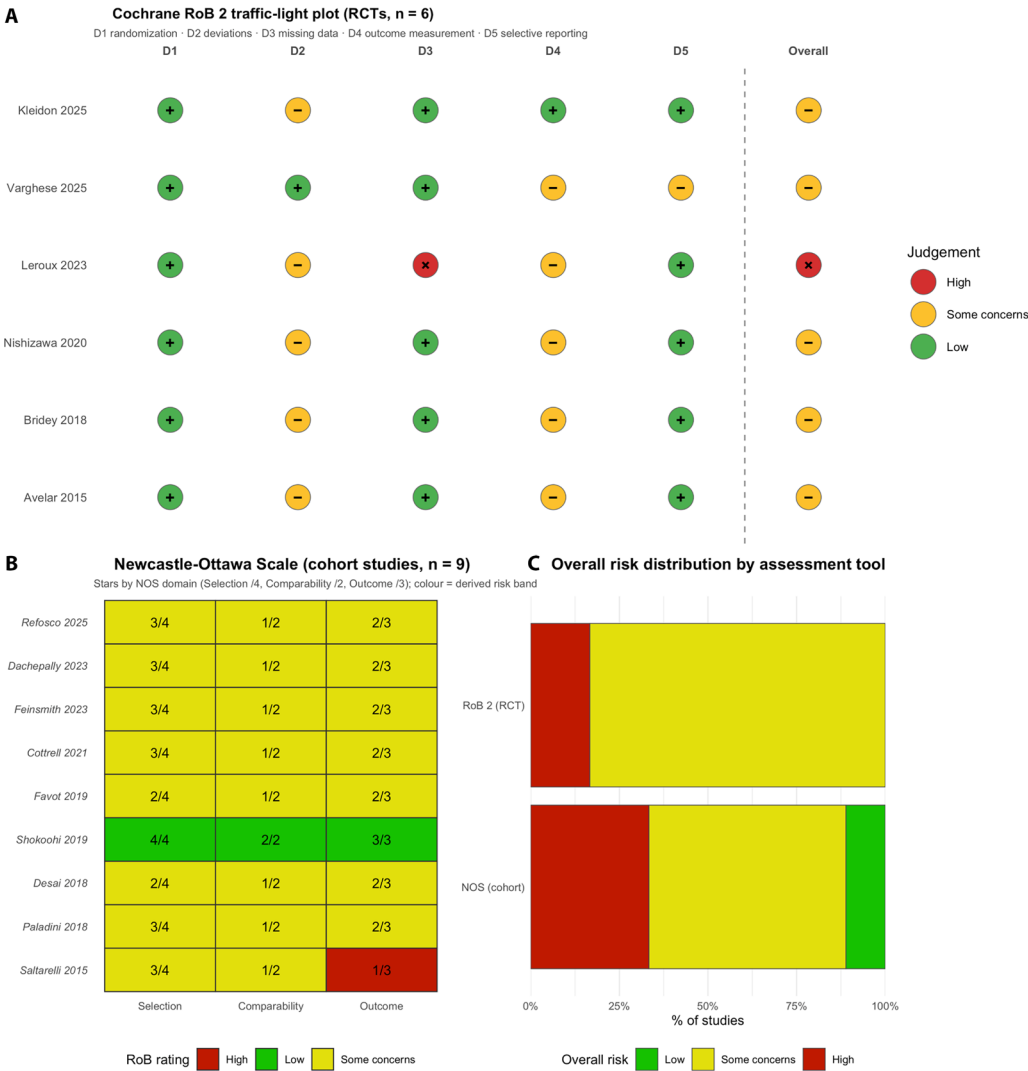


Figure 1. Risk-of-bias and methodological-quality summary across all 15 included studies (Based on Cochrane RoB 2 for 6 RCTs and Newcastle-Ottawa Scale for 9 cohort studies; see Table 2 for details) (A) Cochrane RoB 2 traffic-light plot (RCTs, n = 6). Uses traffic-light colors (green/yellow/red) to display the risk-of-bias judgments across domains for each of the 6 RCTs. (B) Newcastle-Ottawa Scale (cohort studies, n = 9). For each of the 9 cohort studies, stars are shown per NOS domain (Selection /4, Comparability /2, Outcome /3); colors indicate the derived risk band. (C) Overall risk distribution by assessment tool. Summarizes the results from both tools, showing the number or proportion of studies rated as low, some concerns, and high risk of bias overall.

records advanced to full-text retrieval and screening; 4 were excluded at the full-text stage (1 single-arm comparator absent, 1 Portuguese-language companion report, 1 unpublished, 1 duplicate/protocol report of the same EPIC trial), and 15 studies met all inclusion criteria and were included in the qualitative synthesis. Of these, subsets contributed to individual meta-analyses depending on outcome reporting and eligibility for pooling (Figure 2; PRISMA flow diagram).

The 15 included studies comprised 6 RCTs and 9 cohort studies, enrolling a total of 78,591 participants (Table 2). Studies were conducted in Australia (n = 1), Brazil (n = 1), France (n = 1), India (n = 1), Italy (n = 2), Japan (n = 1), and the United States (n = 8). Seven studies enrolled adult participants and eight enrolled pediatric patients. Settings included the emergency department (n = 8), intensive care unit (n = 3), inpatient hospital setting (n = 2), a quaternary

Table 1. GRADE summary of findings for post-insertion outcomes of ultrasound-guided versus landmark peripheral intravenous catheters.

Outcome	No. of studies (design)	Participants	Effect (USG vs landmark)	Certainty (GRADE)	Reasons for downgrading
Catheter failure	4 (2 RCTs, 2 cohorts)	3,404	RR 1.23 (95% CI 1.00–1.51)	⊕⊕○○ Low	Risk of bias; imprecision
Dwell time	3 (2 RCTs, 1 cohort)	929	Not pooled ($I^2 = 91.9\%$); individual estimates conflicting	⊕○○○ Very low	Inconsistency (extreme); risk of bias; imprecision
Infiltration	3 (2 RCTs, 1 cohort)	3,117	RR 1.30 (95% CI 0.82–2.05)	⊕○○○ Very low	Risk of bias (incl. differential ascertainment in 1 RCT); imprecision
Extravasation	3 (2 RCTs, 1 cohort)	29,682	Not pooled ($I^2 = 95.7\%$); directionally inconsistent	⊕○○○ Very low	Inconsistency (extreme); risk of bias; indirectness
Time-to-event-adjacent	2 (2 cohorts)	43,800	Not pooled — Feinsmith adjusted HR 0.91 (0.87–0.95); Shokoohi 72-h RR 1.26 (0.88–1.80)	⊕○○○ Very low	Different effect measures (HR vs RR); risk of bias; conflicting directions; imprecision

RR > 1 indicates a higher event rate (worse outcome) with ultrasound guidance. GRADE certainty: ⊕⊕⊕⊕ High, ⊕⊕⊕○ Moderate, ⊕⊕○○ Low, ⊕○○○ Very low. Certainty was rated per outcome using the GRADE approach (risk of bias, inconsistency, indirectness, imprecision, publication bias); observational cohorts started at low certainty. Publication bias could not be formally assessed (fewer than 10 studies per outcome).

children's hospital (n = 1), and a pediatric surgical unit (n = 1) (Table 2). Six studies specifically enrolled DIVA populations, while the remainder included patients of mixed or all difficulty levels (Table 2).

Full study characteristics are detailed in Table 2. Key features are summarized below. Among the six RCTs, Kleidon et al. (2025) conducted the EPIC trial in 164 pediatric patients (Australia) (14); Varghese et al. (2025) enrolled 90 pediatric ED patients with two-hour follow-up only (India) (21); Leroux et al. (2023) enrolled 223 adult ED patients (222 PIVCs analyzed) but was substantially underpowered (target 582; USA) (22); Nishizawa et al. (2020) randomized 60 adult ICU DIVA patients to nurse-performed USG versus standard care (Japan) (16), reporting post-insertion extravasation as a secondary outcome (13.6% vs 28.5%, $p=0.394$); and Bridey et al. (2018) randomized 114 adult ICU DIVA patients (France) (23). Avelar et al. (2015) randomized 382 venipunctures in 335 pediatric surgical patients (Brazil) using identical 19-mm catheters in both arms, reporting infiltration and phlebitis as post-insertion outcomes (5).

Among the nine cohort studies, Feinsmith et al. (2023) was the largest (n=43,470 adult DIVA patients; USA) (24). Cottrell et al. (2021) is notable as the only unconfounded study reporting significantly

longer USG dwell time (mean 96.1 vs. 59.4 hours; $p<0.001$) using identical catheter types in both groups, though USG cases were disproportionately classified as difficult (75% vs. 36%) (25). Shokoohi et al. (2019) prospectively followed 330 adult ED patients with 72-hour follow-up, achieving the highest NOS score (9/9) (26).

Four cohort studies were identified as having major catheter-specification confounding (detailed in the section below): Refosco et al. (2025) (27), Dachepally et al. (2023) (28), Desai et al. (2018) (15), and Paladini et al. (2018) (29). In all four, USG catheters were systematically longer or of different material than comparator catheters, rendering outcome differences unattributable to ultrasound guidance alone.

Risk of bias

Among the six RCTs, five were judged to have “some concerns” overall (Kleidon et al., 2025 (14); Varghese et al., 2025 (21); Nishizawa et al., 2020 (16); Bridey et al., 2018 (23); Avelar et al., 2015 (5)), primarily related to performance bias inherent in open-label designs where blinding of the operator is not feasible; for Avelar, complications in the USG arm were ascertained with ultrasound visualization whereas

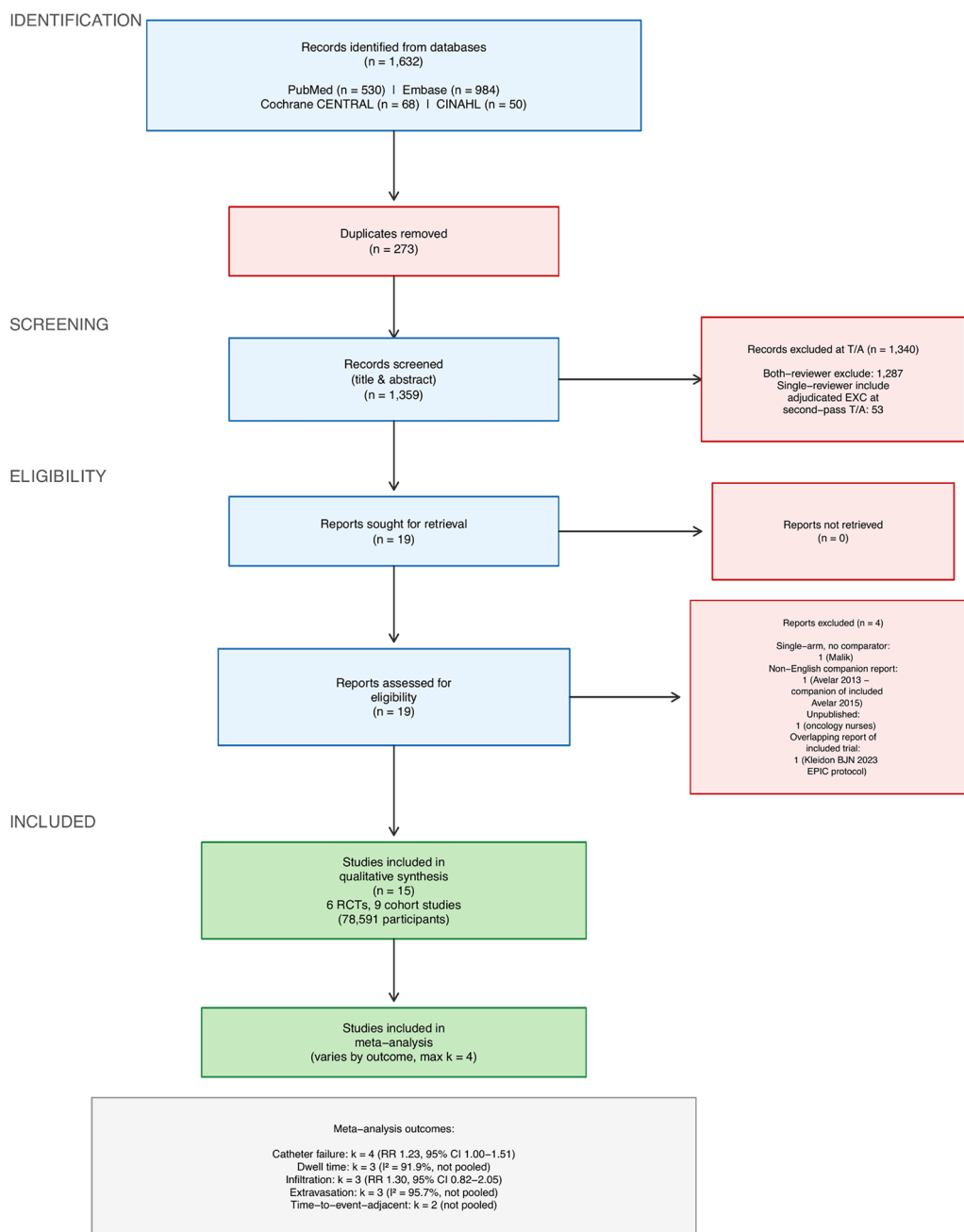


Figure 2. PRISMA 2020 Flow Diagram.

the landmark arm was assessed clinically, a differential outcome ascertainment that may bias the infiltration comparison. Leroux et al. (2023) was rated as high risk of bias due to substantial attrition and failure to reach the target sample size (22).

Among the nine cohort studies, NOS scores ranged from 5 to 9 (Table 2). Shokoohi et al. (2019) achieved the maximum score of 9/9 (26). Refosco et

al. (2025) (27), Dachepally et al. (2023) (28), Fein-smith et al. (2023) (24), Cottrell et al. (2021) (25), and Paladini et al. (2018) (29) each scored 6/9, with points deducted primarily in the comparability domain (cath-eter type confounding for Dachepally, Paladini, and Refosco; difficulty-classification imbalance for Cot-trell). Favot et al. (2019) (30), Desai et al. (2018) (15), and Saltarelli et al. (2015) (31) scored 5/9 (Table 3).

Table 2. Characteristics of included studies.

Study	Year	Country	Design	Setting	Population	N (USG/LM)	DIVA Only	Catheter Confounded	Follow-up	Primary Post-insertion Outcome	Key Finding
Kleidon TM (14)	2025	Australia	RCT	Quaternary children's hospital	Pediatric 0-18y	84/80	No	No	Until removal	PIVC failure, dwell time	Failure 23.8% vs 25.0% (NS); Dwell 47.1 vs 47.7h (NS)
Refosco M (27)	2025	Italy	Retrospective cohort	Pediatric ED	Pediatric 0-18y	135/135	No	Yes*	Until removal	Dwell time, complications	Dwell 5.3±4.0 vs 2.5±1.8 days (p<0.001); Complications 34% vs 70%
Varghese S (21)	2025	India	RCT	Tertiary pediatric ED	Pediatric 1mo-17y	45/45	Mixed	No	2 hours	Catheter patency, complications	Patency 97.78% vs 84.44% (p=0.03)
Dachepally R (28)	2023	USA	Retrospective cohort	Pediatric ICU	Pediatric ICU (DIVA)	99/88	Yes	Yes*	Until removal	Dwell time, IV failure	Median 219 vs 108h (p<0.001); HR 2.20 (1.45-3.34)
Feinsmith SE (24)	2023	USA	Retrospective cohort	Tertiary academic center	Adult (DIVA)	7,972/35,498	Yes	No	30 days	KM survival, HR	Adjusted HR 0.91 (0.87-0.95) favoring USG
Leroux S (22)	2023	USA	RCT	Tertiary ED	Adult ED	106/116	No	No	48-96h	Postinsertion failure, utility time	Failure 11.3% vs 9.5% (NS); Utility 23.5 vs 24.7h (NS)
Cottrell JT (25)	2021	USA	Retrospective cohort	Pediatric hospital	Pediatric (all)	256/287	No	No	Until removal	Dwell time	Mean 96.1 vs 59.4h (p<0.001)
Nishizawa T (16)	2020	Japan	RCT	Mixed ICU	Adult ICU (DIVA)	30/30	Yes	No	Until removal	First-attempt success, extravasation	First-attempt 70% vs 40% (p<0.05); Extravasation 13.6% vs 28.5% (NS)
Favot M (30)	2019	USA	Retrospective cohort	Tertiary ED	Adult ED (CT contrast)	291/29,217	No	No	33 months	Contrast extravasation	4.1% vs 0.21% (RR 19.4, 10.6-35.6)
Shokoohi H (26)	2019	USA	Prospective cohort	Tertiary ED	Adult ED (all access)	182/148	No	No	72 hours	Premature removal	27% vs 21% (RR 1.26, 0.88-1.80, NS)

Study	Year	Country	Design	Setting	Population	N(USG/LM)	DIVA Only	Catheter Confounded	Follow-up	Primary Post-insertion Outcome	Key Finding
Bridey C (23)	2018	France	RCT	Tertiary ICU	Adult ICU (DIVA)	57/57	Yes	No	Max 3 days	Catheter lifespan, extravasation	Lifespan 3 vs 3 days (NS); Extravasation 34% vs 18% (p=0.094)
Desai K (15)	2018	USA	Retrospective cohort	Pediatric ED	Pediatric ED	160/363	No	Yes*	Until removal	Dwell time, complications	KM median 143 vs 89h (p<0.001); Complications 34% vs 32% (NS)
Paladini A (29)	2018	Italy	Prospective cohort	Pediatric ED	Pediatric ED (DIVA) >10y	20/20	Yes	Yes*	Until removal	Dwell time, complications	Dwell 9.2±6.0 vs 3.2±2.1 days (p<0.0001)
Avelar AF (5)	2015	Brazil	RCT	Pediatric surgical unit	Pediatric (12 d-17 y)	188/194	No	No	To catheter removal	Infiltration, phlebitis	Higher infiltration with USG (34/161 vs 23/178; P=.025); phlebitis NS (2/161 vs 7/178); identical 19-mm catheters both arms
Saltarelli NA (31)	2015	USA	Retrospective cohort	Tertiary ED	Adult ED (DIVA)	885/1,803	Yes	No	Hospital stay	Infiltration	7.6% vs 5.9% (OR 1.31, 0.92-1.88, NS)

*USG group used systematically longer catheters than landmark group; results confounded by catheter length/type.

Avelar 2015 randomized 382 venipunctures (188 ultrasound-guided / 194 landmark) in 335 children; post-insertion complications were analyzed among the 339 catheters successfully placed (161 ultrasound-guided / 178 landmark). The 188/194 venipuncture counts are shown here and in the participant total; the 161/178 catheter counts are the denominators used in the infiltration meta-analysis.

Table 3. Risk of bias assessment.

Part A: Cochrane RoB 2 for Randomized Controlled Trials						
Study	D1: Randomization	D2: Deviations	D3: Missing data	D4: Measurement	D5: Reporting	Overall
Kleidon 2025 (14)	Low	Some concerns	Low	Low	Low	Some concerns
Varghese 2025 (21)	Low	Low	Low	Some concerns	Some concerns	Some concerns
Leroux 2023 (22)	Low	Some concerns	High	Some concerns	Low	High risk
Nishizawa 2020 (16)	Low	Some concerns	Low	Some concerns	Low	Some concerns
Bridey 2018 (23)	Low	Some concerns	Low	Some concerns	Low	Some concerns
Avelar 2015 (5)	Low	Some concerns	Low	Some concerns	Low	Some concerns
Part B: Newcastle-Ottawa Scale for Cohort Studies						
Study	Selection (max 4)	Comparability (max 2)	Outcome (max 3)	Total	Key Concern	
Refosco 2025 (27)	3	1	2	6/9	Catheter type confounding	
Dachepally 2023 (28)	3	1	2	6/9	Catheter length confounding	
Feinsmith 2023 (24)	3	1	2	6/9	Selection bias (DIVA), missing IFR data	
Cottrell 2021 (25)	3	1	2	6/9	Retrospective; different difficulty levels	
Favot 2019 (30)	2	1	2	5/9	Ecological comparison, different data sources	
Shokoohi 2019 (26)	4	2	3	9/9	(Highest quality)	
Desai 2018 (15)	2	1	2	5/9	USG group 55% DIVA hx vs TPIV 7.2%; longer catheters	
Paladini 2018 (29)	3	1	2	6/9	Catheter type confounding	
Saltarelli 2015 (31)	3	1	1	5/9	Conference abstract only	

Meta-Analysis results

PRIMARY OUTCOME: CATHETER FAILURE

Four studies reported catheter failure or premature removal as a dichotomous outcome and were eligible for pooling (Kleidon et al., 2025 (14); Leroux et al., 2023 (22); Shokoohi et al., 2019 (26); Saltarelli et al., 2015 (31)). The pooled random-effects RR was 1.23 (95% CI 1.00–1.51; $p = 0.056$; $I^2 = 0.0\%$), indicating no statistically significant difference between USG and landmark groups (Figure 3). Although the lower bound rounds to 1.00 at two decimal places, the

precise value (0.995) confirms that the confidence interval crossed unity. The point estimate favored landmark insertion. In a post-hoc sensitivity analysis using the REML estimator and HKSJ confidence interval adjustment (not pre-specified in PROSPERO), the result was directionally consistent (RR 1.23, 95% CI 1.00–1.51; $p = 0.047$). The primary DL estimate ($p = 0.056$) and HKSJ sensitivity estimate ($p = 0.047$) both straddle the conventional $\alpha = 0.05$ threshold, indicating that statistical significance is borderline and sensitive to estimator choice; clinical interpretation should therefore emphasize the point estimate and the overall

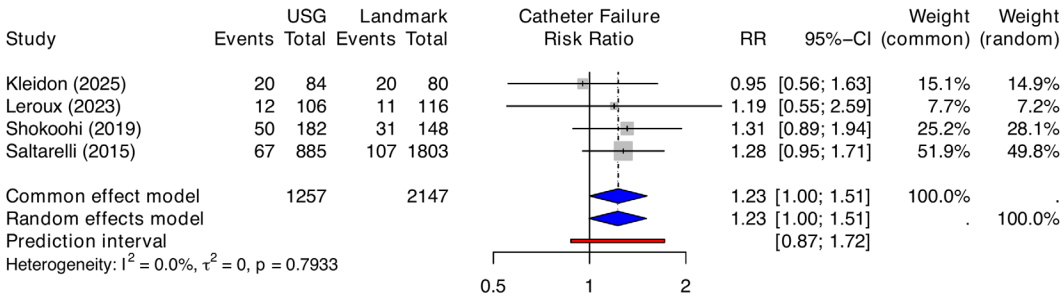


Figure 3. Forest plot: catheter failure or premature removal, USG vs landmark PIVCs ($k = 4$: 2 RCTs + 2 cohort studies). Random-effects (DerSimonian-Laird) pooled estimate shown; the borderline signal (RR 1.23, 95% CI 1.00–1.51, $p = 0.056$) is driven primarily by the cohort component (see Results, Catheter Failure). PIVC type and intravascular catheter-segment length are reported in Table 1 where available; standard short PIVCs predominate in the four pooled studies.

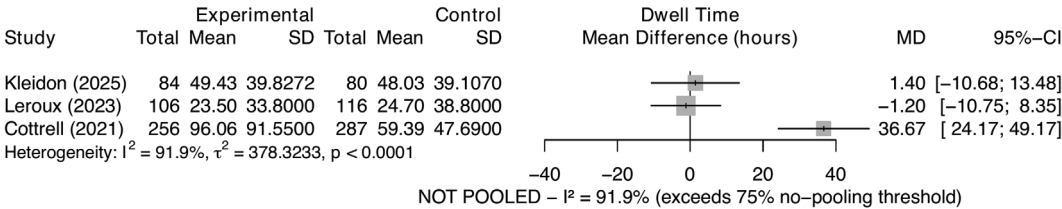


Figure 4. Forest plot: catheter dwell time (USG vs landmark), continuous outcome ($k = 3$: 2 RCTs + 1 cohort). The pooled diamond is intentionally suppressed under the pre-specified $I^2 > 75\%$ rule ($I^2 = 91.9\%$); only individual study estimates are shown. Cottrell 2021 used identical catheter types in both groups but disproportionately classified USG cases as difficult; the other two unconfounded studies (Kleidon 2025; Leroux 2023) used standard short PIVCs and showed no dwell-time difference. Studies in which the USG arm received systematically longer catheters than the landmark arm (Refosco 2025; Desai 2018; Dachepally 2023; Paladini 2018) are reported in “Studies with major confounding” rather than in this forest plot.

pattern of evidence rather than significance alone. A fourth cohort study (Desai et al. 2018) (15) was eligible by outcome but excluded from primary pooling due to selection bias and catheter length confounding. Leave-one-out sensitivity analysis demonstrated that removing Kleidon et al. (2025), the only pediatric study in this pool, shifted the estimate to RR 1.28 (95% CI 1.02–1.60; $p = 0.032$), while removing Saltarelli et al. (2015) (26), the largest contributor at 49.8% random-effects weight, attenuated the estimate to RR 1.18 (95% CI 0.88–1.58; $p = 0.28$); the borderline catheter-failure result therefore depends substantially on the largest single study (see Supplementary Material S4: leave-one-out for catheter failure). This borderline signal also arose predominantly from the cohort studies; the RCT-only estimate was null (RR 1.03, 95% CI 0.66–1.61), as detailed in the design-stratified analysis below.

A post-hoc design-stratified analysis showed that the catheter-failure signal arose primarily from the

cohort studies ($k = 2$; RR 1.29, 95% CI 1.02–1.63; $p = 0.036$), whereas the RCT-only estimate was essentially flat ($k = 2$; RR 1.03, 95% CI 0.66–1.61; $p = 0.91$). The design-by-effect interaction test was non-significant ($p = 0.37$); with only 2 studies in each design tier this comparison is severely underpowered. This pattern suggests that the overall estimate is influenced more by the cohort studies than by the RCTs.

PRIMARY OUTCOME: DWELL TIME

Three studies reported dwell time in formats suitable for pooling (Kleidon et al., 2025; Leroux et al., 2023; Cottrell et al., 2021) (22, 25, 32). However, pooling was precluded by substantial heterogeneity ($I^2 = 91.9\%$; Figure 4). Kleidon et al. reported no difference (median 47.1 vs. 47.7 hours), Leroux et al. similarly found no difference (mean 23.5 vs. 24.7 hours; $p = 0.808$), while Cottrell et al. reported a significant 36.7-hour advantage

favoring USG (mean 96.06 vs. 59.39 hours; $p < 0.001$) (Figure 4). The conflicting results likely reflect differences in populations, operator expertise, and clinical context: Cottrell et al. studied a specialized vascular access team with extensive USG experience in a pediatric setting, where USG may confer particular advantages due to the unique challenges of pediatric venous access. Narrative synthesis of these three unconfounded studies suggests that any dwell time benefit of USG may be setting-dependent and operator-dependent rather than a generalizable effect of the guidance modality. Two additional studies reporting dwell time (Refosco 2025, Desai 2018) (15, 27) were excluded from this analysis due to catheter length confounding.

SECONDARY OUTCOME: INFILTRATION

Three studies reported infiltration rates (Varghese et al., 2025 (21); Saltarelli et al., 2015 (31) Avelar et al., 2015 (5)). The pooled RR was 1.30 (95% CI 0.82–2.05; $p = 0.26$; $I^2 = 48.8\%$), indicating no significant difference (Supplementary Figure 1). A sensitivity analysis using the REML estimator with Hartung-Knapp-Sidik-Jonkman adjustment was consistent (RR 1.33, 95% CI 0.62–2.85; $p = 0.25$). Varghese et al. found a non-significant trend favoring USG (2.22% vs. 11.11%; $p = 0.09$) (21), while Saltarelli et al. reported a non-significant trend against USG (7.6% vs. 5.9%; OR 1.31, 95% CI 0.92–1.88) (31); the newly included Avelar trial reported significantly higher infiltration with USG (34/161 [21.1%] vs. 23/178 [12.9%]; $P = .025$) (5), although complications in its USG arm were ascertained with ultrasound visualization, which may have increased detection (Supplementary Figure 1).

SECONDARY OUTCOME: EXTRAVASATION

Three studies reported extravasation events (Nishizawa et al. 2020, Favot et al. 2019, Bridey et al. 2018) (16, 23, 30); however, the fundamental incompatibility of their populations and outcome definitions precludes any meaningful pooled inference, and individual study results are the primary informative unit. Nishizawa et al. reported a directionally opposite finding (13.6% vs 28.5%, $p = 0.394$) in a small ICU RCT (Supplementary Figure 2) (16). Favot et al. reported

contrast extravasation rates of 4.1% with USG versus 0.21% with standard PIVCs (RR 19.4, 95% CI 10.6–35.6), confined to a highly specific high-pressure CT contrast injection context (Supplementary figure 1) (30). Bridey et al. found extravasation as a reason for catheter removal in 34% of USG versus 18% of landmark catheters ($p = 0.094$) in an adult ICU setting (Supplementary Figure 2) (23). In keeping with the pre-specified $I^2 > 75\%$ rule, these three studies were not pooled ($I^2 = 95.7\%$) and no summary estimate is reported (Supplementary Figure 2).

TIME-TO-EVENT-ADJACENT ESTIMATES (NOT POOLED)

Two studies reported effect estimates with a time-to-event component but using different statistical metrics, precluding meaningful meta-analytic pooling. Feinsmith et al. (2023) (24) reported an adjusted hazard ratio (HR) of 0.91 (95% CI 0.87–0.95) favoring USG for catheter survival at 30 days, derived from a multivariable Cox proportional hazards model (Supplementary Figure 3). Shokoohi et al. (2019) (26) reported a premature removal risk ratio (RR) of 1.26 (95% CI 0.88–1.80) trending against USG at a fixed 72-hour follow-up window, a cumulative-incidence ratio rather than a hazard ratio (Supplementary Figure 3). Because HR and fixed-window RR are conceptually different effect measures (the HR conditions on time-at-risk; the RR does not), an attempted pooling on the log-HR scale would mix incompatible quantities and risk producing a spuriously precise summary; we therefore present each study's estimate individually without a pooled summary statistic. The divergent directions of the 30-day adjusted HR and the 72-hour cumulative-incidence RR most plausibly reflect their different time windows and effect metrics rather than a true difference in catheter performance; the corresponding forest plot has been relocated to Supplementary Figure 3 (24, 26).

Subgroup and sensitivity analyses

ADULT VS. PEDIATRIC SUBGROUP (CATHETER FAILURE)

Among adult studies ($k = 3$), the pooled RR was 1.28 (95% CI 1.02–1.60; $p = 0.03$), with the point

estimate trending unfavorable to USG (Supplementary Figure 4). Only one pediatric study (Kleidon et al., 2025) (14) contributed to this outcome, with RR 0.95 suggesting no difference (Supplementary Figure 4). The test for subgroup differences was not significant ($p = 0.32$), indicating that the observed adult-subgroup result should be interpreted as hypothesis-generating rather than confirmatory (Supplementary Figure 4). (See “Primary Outcome: Catheter Failure” above for the post-hoc design-stratified analysis [RCT-only vs cohort-only] and the leave-one-out sensitivity analysis showing dependence on the largest cohort study.)

STUDIES WITH MAJOR CONFOUNDING

Four studies were excluded from the primary pooled analyses due to systematic differences in catheter specifications between groups. Refosco et al. (2025) reported longer dwell times (5.3 ± 4.0 vs. 2.5 ± 1.8 days; $p < 0.001$) and fewer complications (34% vs. 70%; $p < 0.001$) with USG in a pediatric ED, but USG cannulas were predominantly 64 mm long peripheral catheters compared with 19–32 mm short cannulas in the blind group (27). Dachepally et al. (2023) reported dramatically longer dwell times for USG catheters (median 219 vs. 108 hours; HR 2.20, 95% CI 1.45–3.34) in a pediatric ICU setting, but USG catheters were exclusively 4.4 cm in length compared with 1.9–3.2 cm in the traditional group (28). Desai et al. (2018) similarly reported longer Kaplan-Meier median survival with USG (143 vs 89 hours, log-rank $p < 0.001$) in a pediatric ED, but USG patients disproportionately had a history of difficult access (55% vs 7.2%) and the authors acknowledged that the majority of USG catheters were longer than traditional catheters (15). Paladini et al. (2018) reported significantly longer dwell times (9.2 ± 6.0 vs. 3.2 ± 2.1 days; $p < 0.0001$) and fewer complications (25% vs. 70%; $p = 0.003$) with USG, but USG catheters were 8 cm polyurethane cannulas placed via Seldinger technique versus short Teflon catheters in the landmark group (29). In all four studies, the observed benefits are more plausibly attributed to greater intravascular catheter length and catheter material, and in Desai, selection bias, rather than to ultrasound guidance per se. In a sensitivity analysis including all studies reporting dwell time (adding Refosco and Desai to the confounded set), heterogeneity remained extreme.

Discussion

Interpretation of main findings

This systematic review and meta-analysis is, to our knowledge, the first to specifically and comprehensively evaluate post-insertion outcomes of USG compared with landmark PIVC placement. Across 15 studies enrolling 78,591 participants, we found no statistically significant evidence that USG insertion improves catheter failure rates, dwell time, or other post-insertion complications. The pooled RR for catheter failure was 1.23 (95% CI 1.00–1.51; $p = 0.056$), a non-significant estimate with a point estimate above 1.0 (Figure 3). Dwell time results were heterogeneous and could not be pooled ($I^2 = 91.9\%$) (Figure 4). These findings challenge the implicit assumption that the well-documented insertion benefits of USG translate into superior catheter performance after placement.

The addition of Cottrell et al. (2021) (25) introduced substantial heterogeneity into the dwell time analysis ($I^2 = 91.9\%$) (Figure 4), preventing meta-analytic pooling. This study’s positive finding, a 36.7-hour dwell time advantage for USG, is notable because it used identical catheter types in both groups, thus controlling for the catheter length confounding present in other positive studies (Figure 4). Cottrell et al.’s findings should be interpreted within the appropriate clinical context, as patients in the ultrasound-guided group were more frequently classified as having difficult vascular access (75% vs. 36%) (25). Additionally, the involvement of a specialized vascular access team may have contributed to improved catheter placement outcomes through expert vessel selection, rather than reflecting the isolated effect of ultrasound guidance itself.

Our results must be interpreted against the broader context of USG PIVC literature. Prior systematic reviews and meta-analyses have consistently demonstrated that USG improves first-attempt success rates and reduces the number of skin punctures, particularly in DIVA populations (4, 6, 7). These insertion benefits are clinically meaningful, as they reduce patient pain and procedural time. However, the post-insertion phase represents the majority of the catheter’s clinical lifecycle, and our findings did not detect

a statistically significant post-insertion advantage for USG over landmark placement once the catheter is successfully sited (although our pre-specified analyses were not designed as a formal equivalence trial and we did not pre-define an equivalence margin). This distinction is important for clinical decision-making: USG should be valued for its ability to achieve venous access in difficult patients, but clinicians should not expect improved catheter longevity or reduced complication rates as a consequence of the guidance technique alone (9, 10).

A critical finding of this review is the identification of catheter length and type as likely confounders in studies reporting favorable USG outcomes. Refosco et al. (2025) (27), Dachepally et al. (2023) (28), Desai et al. (2018) (15), and Paladini et al. (2018) (29) demonstrated markedly superior outcomes with USG, but in all these studies, USG catheters were substantially longer than those in the comparator group. The addition of Nishizawa et al. (2020) (16), an RCT in critically ill patients, supports the absence of meaningful post-insertion benefits even in a high-acuity DIVA population, while Desai et al. (2018) (15) once again illustrates how catheter length confounding can produce apparent dwell-time benefits that do not survive methodological scrutiny. This confounding makes it impossible to attribute the observed benefits to ultrasound guidance rather than to the greater intravascular catheter length, which has been independently associated with improved catheter survival (32, 33).

An important question arising from this observation is whether USG inherently induces the use of longer catheters. By visualizing deeper, non-palpable veins – typically the brachial, basilic, or deep forearm veins at depths of 1.0 to 2.0 cm – USG cannulation often necessitates longer catheters to ensure an adequate intravascular segment. When a standard 4.5 cm catheter is inserted into a vessel at 1.5 cm depth at a 45-degree angle, the resulting intravascular length may be as little as 1.5 to 2.0 cm, leaving a substantial portion of the catheter in subcutaneous tissue (33). Pandurangadu et al. demonstrated that 100% of USG catheters failed when less than 30% of the catheter length resided within the vein (33). In this context, the superior outcomes observed in studies using longer catheters (4.4 cm in Dachepally et al. (28); 8 cm

in Paladini et al. (29)) may reflect the physical requirement of matching catheter length to vessel depth rather than an independent benefit of ultrasound guidance per se. This distinction has profound clinical implications: the benefit may not lie in “seeing the vein” during insertion, but rather in selecting an appropriately long catheter for the depth of the target vessel. The catheter-to-vein ratio, which van Loon et al. identified as a critical determinant of cannulation success with an optimal cut-off of approximately 0.41, may be equally important for post-insertion catheter survival (32). Future studies must therefore control for catheter specifications, including length, gauge, and material, to isolate the independent effect of the ultrasound guidance modality from the confounding effect of catheter selection.

Extravasation as a potential safety signal

An exploratory extravasation signal emerges from three studies that warrants acknowledgment, while noting that the certainty of this evidence is very low (GRADE ⊕○○○). Bridey et al. (2018) (23) and Favot et al. (2019) (30) reported numerically higher extravasation rates with USG, though the magnitude differed dramatically (RR 1.92 vs. 19.4); Nishizawa et al. (2020) (16) reported a directionally opposite finding (13.6% vs. 28.5%). The extreme between-study heterogeneity ($I^2 = 95.7\%$) precluded meta-analytic pooling, and the directional inconsistency together with heterogeneity of outcome definitions means the extravasation data are best interpreted as hypothesis-generating, requiring confirmation in adequately powered prospective studies before clinical inferences can be drawn.

The underlying biomechanics offer a plausible explanation. USG enables cannulation of deeper, non-palpable veins, typically at depths of 1.0 to 2.0 cm below the skin surface (9). When standard-length catheters (4.5 cm) are used at these depths, the angle of insertion required to reach the vessel results in a shorter intravascular catheter segment. Fields et al. demonstrated that vessel depth is inversely associated with USG catheter longevity, with deeper insertions at greater risk of failure (9). The short intravascular segment is susceptible to catheter tip migration with patient movement, particularly at flexion points such as

the antecubital fossa. When the catheter tip migrates outside the vessel lumen, any infusate, but especially high-pressure contrast injections, can extravasate into the perivascular tissue.

The dramatically elevated Favot et al (30). RR (19.4) for contrast extravasation likely reflects standard-length catheter vulnerability compounded by high-pressure injection during CT contrast administration; Bridey et al (23). observed a more modest but still concerning trend (34% vs 18%, $p = 0.094$) in an ICU setting where USG catheters were also larger gauge (18G vs 21G, $p = 0.059$), which may have contributed independently to vessel wall trauma.

If this directionally inconsistent extravasation pattern is confirmed in adequately powered prospective studies, clinical practice may need to consider using appropriately longer catheters (e.g., 4.78 cm or longer) when USG is used to access deep veins, and particular caution may be warranted before using USG-placed standard-length PIVCs for high-pressure contrast injection. We frame this only as a hypothesis for prospective evaluation; the current evidence is too inconsistent to support a confirmed safety signal.

The subgroup analysis by age group generated a hypothesis worth noting in the context of future research. Among adult studies ($k = 3$), the pooled RR was 1.28 (95% CI 1.02–1.60; $p = 0.03$) (Supplementary Figure 4), with the point estimate trending unfavorably for USG. A biomechanically plausible explanation exists: USG targets deeper veins that may be more prone to catheter tip migration with standard-length catheters. In contrast, the single pediatric study (Kleidon et al., 2025) (14) showed no difference (RR 0.95). However, the subgroup interaction test was not significant ($p = 0.32$), the adult subgroup includes a conference abstract (Saltarelli et al., 2015) (31) with limited methodological detail, and overall GRADE certainty is low. This result should therefore be characterized as an exploratory signal requiring confirmation in adequately powered prospective studies, rather than as evidence of harm.

As reported above (Results, Catheter Failure), the post-hoc design-stratified analysis showed that the borderline overall failure signal arose from the cohort studies (RR 1.29, 95% CI 1.02–1.63) while the RCT-only estimate was flat (RR 1.03, 95% CI 0.66–1.61).

The Feinsmith et al. (2023) study (24), while the largest cohort by sample size, is inherently subject to selection bias: patients who receive USG PIVCs are those who have already failed landmark attempts, representing a population with intrinsically more difficult venous access and potentially different catheter outcomes, illustrating one source of the divergence between cohort-driven and RCT-driven estimates.

Study limitations

This review has several important limitations:

1. The evidence base is small and heterogeneous, comprising only six RCTs and nine cohort studies with varied populations, settings, outcome definitions, and follow-up durations.
2. Inter-rater agreement for title/abstract screening was fair (Cohen's kappa = 0.330; percent agreement 95.8%), reflecting the κ paradox under high exclusion prevalence (34) rather than substantive disagreement. The 57 single-reviewer-include records were adjudicated by the primary reviewer in a pre-specified second-pass review against the PROSPERO PICO, with documented categorical exclusion reasons. Selection-bias risk was mitigated by cross-referencing against three prior USG-PIVC systematic reviews (4, 6, 7) (no eligible studies missed) and an audit trail of all second-pass decisions.
3. Four studies had major confounding by catheter type or selection, necessitating their exclusion from pooled analyses.
4. Saltarelli et al. (2015) (31) was available only as a conference abstract with limited methodological detail.
5. Varghese et al. (2025) (21) employed only a two-hour follow-up, which is insufficient to capture most post-insertion complications.
6. Leroux et al. (2023) (22) was substantially underpowered due to COVID-19-related recruitment challenges.
7. Formal assessment of publication bias was not possible because fewer than 10 studies contributed to any single meta-analysis.

8. The search covered four databases (PubMed, Embase, Cochrane CENTRAL, CINAHL) but did not include Scopus or Web of Science; although three prior systematic reviews were cross-referenced and the reference lists of included studies were hand-searched without identifying additional eligible records, studies uniquely indexed in those databases may have been missed.
9. In the newly included Avelar trial, post-insertion complications in the ultrasound arm were ascertained with ultrasound visualization while the landmark arm was assessed clinically, a differential ascertainment that may inflate the infiltration signal.
10. The overall certainty of evidence as assessed by the GRADE framework ranged from low to very low across all outcomes (Table 1) (17), reflecting the predominance of observational studies, serious imprecision, and the limitations of the available RCTs.

Clinical implications

Clinicians should continue to value USG as a highly useful insertion aid, particularly in patients with difficult intravenous access. However, current evidence does not justify assuming that USG alone improves catheter longevity or reduces post-insertion complications. If post-insertion performance is a clinical priority, attention should focus on catheter selection, including length, gauge, material, and the depth of the target vein, rather than on guidance modality alone.

Overall, the current literature more strongly supports the insertion-phase advantage of USG than any consistent post-insertion advantage. The most plausible interpretation of the available evidence is that post-insertion outcomes are shaped at least as much by catheter specification and study context as by ultrasound guidance itself. Future adequately powered trials that standardize catheter type across groups are needed before the independent post-insertion effect of USG can be determined.

Conclusion

This systematic review and meta-analysis of 15 studies did not demonstrate a consistent post-insertion advantage of ultrasound-guided over landmark-guided PIVC placement across catheter failure, dwell time, infiltration, or extravasation; the current evidence base is too small and heterogeneous to support either firm benefit or firm equivalence claims, and a clinically meaningful effect in either direction cannot be excluded. The pooled catheter failure rate was not significantly different between groups (RR 1.23, 95% CI 1.00–1.51; $p = 0.056$). Dwell time results were heterogeneous ($I^2 = 91.9\%$) and could not be pooled, with one study showing significant benefit and two showing no difference. In adult patients specifically, the pooled RR for catheter failure trended unfavorably for USG (RR 1.28, 95% CI 1.02–1.60; $p = 0.03$), though the subgroup interaction test was non-significant ($p = 0.32$) and this result is best considered hypothesis-generating given the small number of studies and low certainty of evidence. This adult subgroup is moreover dominated by a single conference abstract (Saltarelli et al., ~50% of the pooled weight), and the leave-one-out analysis showed that removing that study rendered the overall catheter-failure estimate non-significant (RR 1.18, 95% CI 0.88–1.58); the adult signal should therefore not be over-interpreted. Extravasation data from two studies (Bridey et al., Favot et al.) (23, 30) were directionally consistent in suggesting a potential signal associated with USG-placed PIVCs, though this was not confirmed by a third study (Nishizawa et al.) (16) and certainty is very low; prospective studies are needed before clinical conclusions can be drawn. Studies reporting strongly favorable USG results were confounded by systematic differences in catheter length and type, highlighting that catheter selection, not the guidance modality, may be the primary determinant of post-insertion catheter performance. This interpretation, and the overall finding of no consistent post-insertion benefit, is conditional on our classification of four catheter-specification-confounded studies and their exclusion from the primary pooled analyses. The current evidence more strongly supports the insertion-phase advantage of USG (well established in

prior systematic reviews (6, 7, 35)) than any consistent post-insertion advantage. Future adequately powered RCTs that control for catheter length, gauge, and material are essential to provide definitive evidence.

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Author's Contributions:

TAL: Acquisition, Conception, Design, Initial Draft, Revision. YLW: Acquisition, Analysis, Initial Draft. JTL: Coordination, Revision. PMC: Revision. CCC: Conception, Acquisition, Revision, Interpretation, Initial Draft. All authors have approved the final version of the manuscript and agree to be accountable for all aspects of the work, ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

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Appendix–Supplementary materials

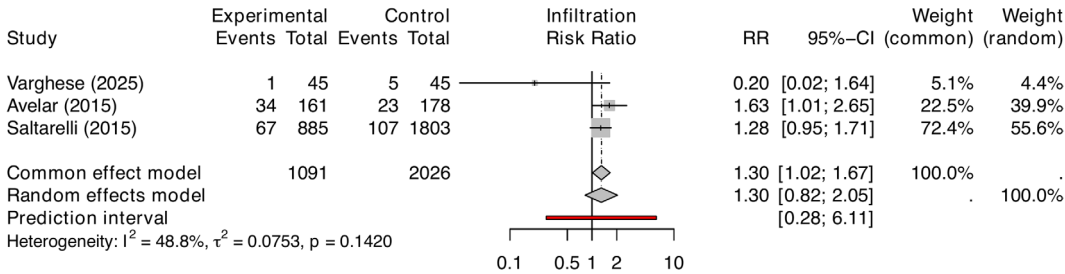


Figure S1. Forest plot: infiltration (USG vs landmark), $k = 3$ (2 RCTs + 1 cohort, all standard-PIVC). Random-effects pooled RR 1.30 (0.82–2.05; $I^2 = 48.8\%$); the newly added Avelar 2015 trial reported higher infiltration with USG.

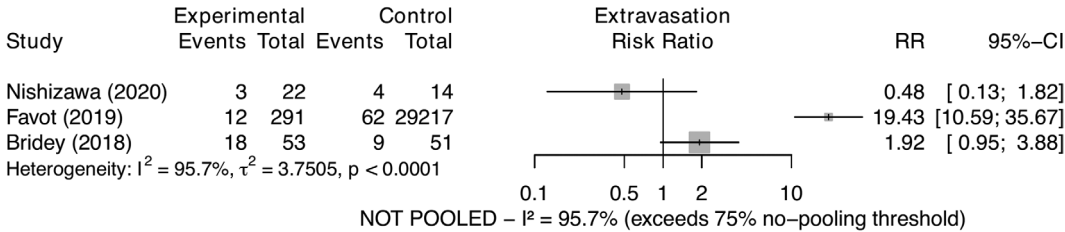


Figure S2. Forest plot: extravasation (USG vs landmark), not pooled ($I^2 = 95.7\%$; Bridey 2018, Favot 2019, Nishizawa 2020). Presented as a supplementary figure because the populations and outcome definitions are statistically incompatible.

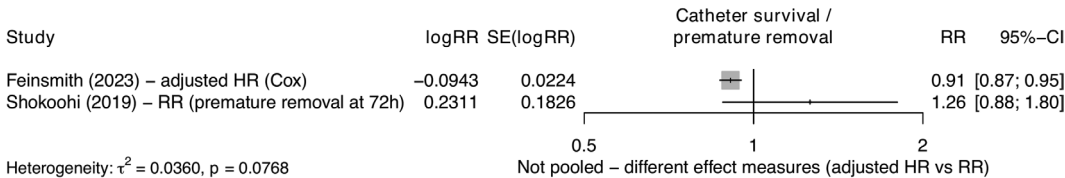


Figure S3. Individual study estimates (NOT pooled) for catheter survival / premature removal: Feinsmith 2023 (adjusted HR from multivariable Cox model, 30-day follow-up) and Shokoohi 2019 (cumulative-incidence RR at fixed 72-hour follow-up). The pooled diamond is intentionally suppressed because HR and cumulative-incidence RR are conceptually different effect measures and pooling them on the log-HR scale would mix incompatible quantities.

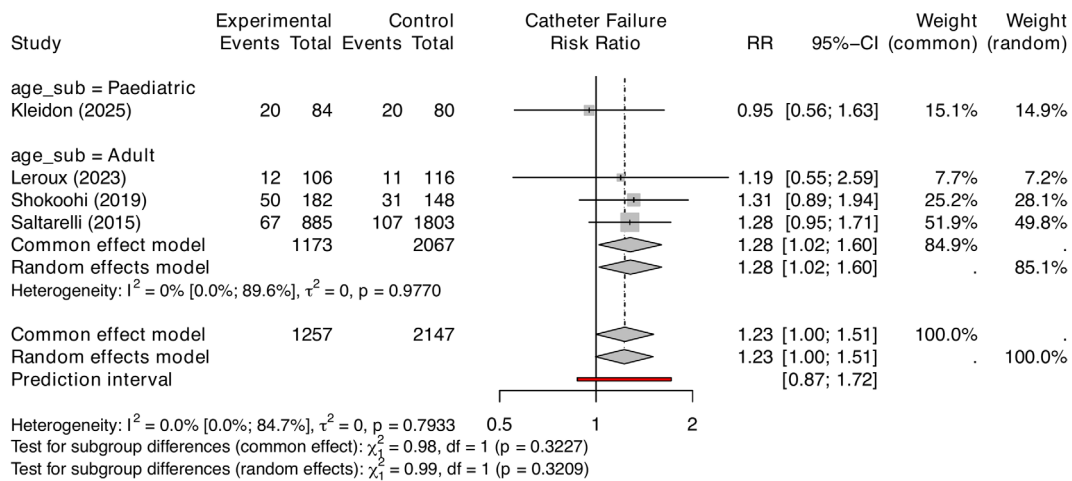


Figure S4. Forest plot: catheter failure subgroup by age (adult vs pediatric, $k = 4$ same as Figure 3). The adult subgroup contains 2 cohorts + 1 conference abstract (Saltarelli 2015 dominates the adult weight); the pediatric subgroup contains a single RCT (Kleidon 2025). Subgroup-by-effect interaction not statistically significant ($p = 0.32$) and severely underpowered.